

AN ENDPOINT REVEAL REFUTES THE EXACT TWO-LIMINAL BURNING FORMULA FOR PATHS

ABSTRACT. The k -liminal burning number $b_k(G)$ of Bonato, Marbach, Marcoux and Mishura interpolates between the cooling number $\text{CL}(G) = b_1(G)$ and the burning number $b(G) = b_{|V(G)|}(G)$: each round a saboteur reveals k vertices and an arsonist burns one revealed vertex, and the saboteur maximises the number of rounds. A recent preprint asserts the exact formula $b_2(P_n) = \lceil (n+2)/3 \rceil$ for every $n \geq 1$. We refute it. The witness is a single two-element set: if the saboteur reveals the two *endpoints* of P_4 in the first round, then three rounds are needed, so $b_2(P_4) = 3 > 2$. The argument is four lines and needs no computation, and the lower bound $b_2(P_4) \geq 3$, which is all the refutation uses, is insensitive to the two clauses of the definition that the sources leave ambiguous and to whether the saboteur must reveal exactly k vertices in later rounds.

1. THE PARAMETER AND THE CLAIM

Let G be a finite simple graph and let $k \geq 1$. In the k -liminal burning game of Bonato, Marbach, Marcoux and Mishura [2], two players alternate on G . In round 1 the saboteur reveals a set of k vertices and the arsonist burns one revealed vertex, the *source*. In each round $i \geq 2$, first every unburned vertex with a burned neighbour burns (*propagation*), then the saboteur reveals k further vertices, then the arsonist burns one revealed unburned vertex if one exists. Revelations are permanent. The game ends in the round in which the last vertex burns; the saboteur maximises the number of rounds and the arsonist minimises it, and $b_k(G)$ is the value under optimal play. Thus $b_1(G) = \text{CL}(G)$, the cooling number, and $b_{|V(G)|}(G) = b(G)$, the burning number. A round in which the last vertex burns by propagation alone is counted; this is forced by the value $b_2(P_2) = 2$ implied by the formula below.

Two clauses of the definition are read differently by the two sources.

- (A) *What may the arsonist burn from round 2 on?* Either any revealed unburned vertex, or only a vertex revealed in the current round.
- (S) *What may the saboteur reveal?* “ k additional vertices” [1] permits re-revealing a vertex that has since burned; [2] does not.

We fix the convention used below: from round 2 on the arsonist may burn any revealed unburned vertex, and the saboteur reveals vertices that are

2020 *Mathematics Subject Classification*. 05C57, 91A43, 05C38.

Key words and phrases. liminal burning number, graph burning, cooling number, path, combinatorial game, counterexample.

neither revealed nor burned. Only the upper bound in Theorem 1 uses either clause, and it uses only the first alternative in (A); the refutation itself does not, as the theorem records.

Ambrose, Angelone, Chen, Ma, Ortiz San Miguel, Watanabe, Whitcomb and Wu [1] state the following.

Theorem 3.1 ([1]). For every $n \geq 1$, $b_2(P_n) = \left\lceil \frac{n+2}{3} \right\rceil$.

2. THE COUNTEREXAMPLE

Write $V(P_4) = \{v_1, v_2, v_3, v_4\}$ with edges v_1v_2, v_2v_3, v_3v_4 .

Theorem 1. $b_2(P_4) = 3$ under the convention fixed in Section 1. The lower bound $b_2(P_4) \geq 3$ holds under either alternative in (A) and in (S), and whether or not the saboteur is required to reveal exactly two vertices in later rounds. Since $\lceil (4+2)/3 \rceil = 2$, Theorem 3.1 of [1] is false.

The object. The witness is the first-round reveal

$$S = \{v_1, v_4\},$$

the two endpoints. Nothing else is needed; in particular no computation is needed.

Proof. Lower bound. Let the saboteur reveal $S = \{v_1, v_4\}$ in round 1. The source must be revealed, hence v_1 or v_4 ; the map $v_i \mapsto v_{5-i}$ is an automorphism of P_4 fixing S , so we may take the source to be v_1 . Suppose the game ended in round 2. At the start of round 2 propagation burns exactly v_2 , so the burned set is $N[v_1] = \{v_1, v_2\}$; the arsonist then burns at most one further vertex. Hence at most $|N[v_1]| + 1 = 3$ of the four vertices are burned at the end of round 2, a contradiction. Therefore $b_2(P_4) \geq 3$.

This step bounds the burned set by a cardinality and never asks *which* vertex the arsonist chose, so it is valid under either alternative in (A) and in (S) — including the loosest, in which the arsonist may burn any unburned vertex at all. It uses only the round-1 reveal, so it is also valid if the saboteur is permitted to reveal fewer than k vertices in later rounds.

Upper bound. Let the saboteur reveal any 2-set S' in round 1. If S' meets $\{v_2, v_3\}$, the arsonist takes such a vertex as source; since $N[v_2] = \{v_1, v_2, v_3\}$ and $N[v_3] = \{v_2, v_3, v_4\}$, propagation alone burns everything by the end of round 3. Otherwise $S' = \{v_1, v_4\}$; the arsonist burns v_1 , and because revelations are permanent, v_4 is still revealed and unburned in round 2, so the arsonist burns v_4 there. The burned set is then $\{v_1, v_2, v_4\}$ and round-3 propagation burns v_3 . In both cases the game ends by round 3, so $b_2(P_4) \leq 3$. This argument uses the first alternative in (A) and is unaffected by whether the round-2 reveal must have exactly two elements or may have fewer. $\square$

Remark. The claimed value 2 requires a source v with $|N[v]| + 1 \geq 4$, i.e. $|N[v]| \geq 3$, i.e. $v \in \{v_2, v_3\}$ — precisely the two vertices the endpoint reveal denies. Correspondingly the proof of Theorem 3.1 in [1] has the saboteur

reveal $\{v_1, v_2\}$ on turn 1, which hands the arsonist the source v_2 , and writes that it “propose[s] a saboteur strategy for $k = 2$ on P_n , and claim[s] that it is optimal” without proving the optimality. Because the saboteur is the *maximiser*, exhibiting one saboteur strategy establishes only a lower bound on $b_2(P_n)$.

3. SCOPE AND WHAT WAS CHECKED

The accompanying program `verify.py` mechanises the argument above on the parsed graph: that the printed reveal is the endpoint set, that the source must then be an endpoint, the round-2 cardinality bound for each possible round-2 burn, and the upper bound over all six 2-subsets of $V(P_4)$. It also computes $b_2(P_4) = 3$ by exact backward induction on the extensive-form game under each of the four combinations of the alternatives in (A) and (S), the saboteur there revealing exactly two vertices whenever two remain available.

No closed form for $b_2(P_n)$ is offered, and nothing is claimed for $n \geq 5$.

REFERENCES

- [1] Ambrose, Angelone, Chen, Ma, Ortiz San Miguel, Watanabe, Whitcomb, and Wu, *Burning games on strong path products*, arXiv:2509.20572v2, 2025.
- [2] Bonato, Marbach, Marcoux, and Mishura, *Between burning and cooling: liminal burning on graphs*, Discrete Math. **349** (2026), 115198. doi:10.1016/j.disc.2026.115198; arXiv:2505.10727.